

Efficient contact deformation model update

Rigid bodies, local material memory and independent torque

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There is a promising cheap improvement for sustained rolling. There is no new validated solution for all experimental collisions. A one-parameter speed-dependent rolling law improves all eight evaluation points in the existing tennis data partition. A related normal-impact law makes the rock sphere-proxy predictions worse; that rock extension is rejected. Both results are retained.

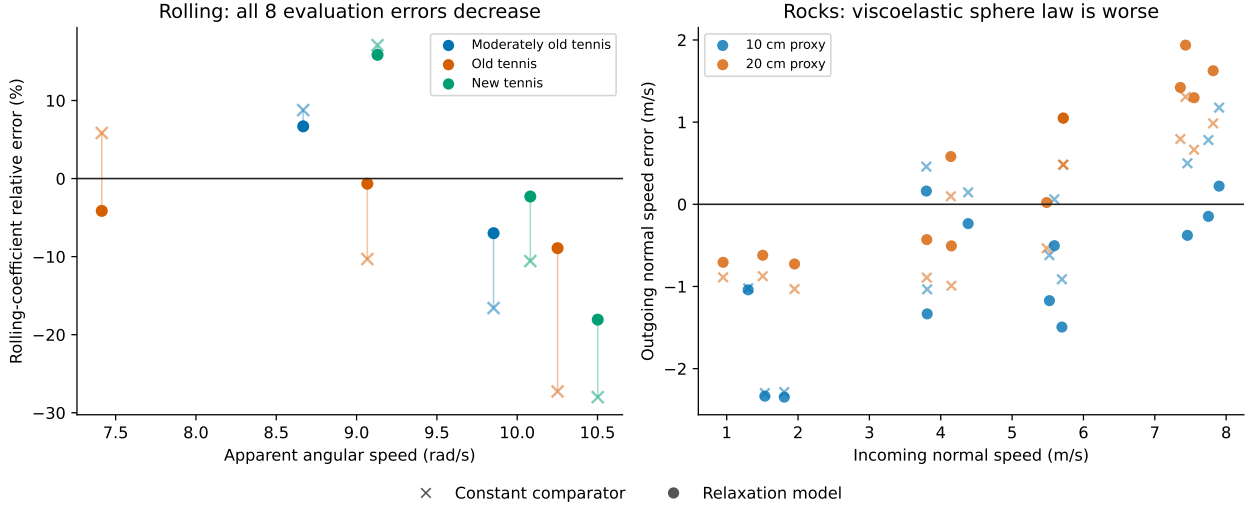
What changed, and what remains fixed

The bodies remain rigid. Local compression and material relaxation represent unresolved deformation. Normal and tangential restitution, static friction μ_s and dynamic friction μ_d already documented for existing profiles remain unchanged. There is no substitution of a sliding coefficient for a measured static coefficient. The new estimates concern missing effective rolling behavior for three tennis specimens. They are not universal rubber material properties.

One nonnegative relaxation parameter per specimen is compared with one nonnegative constant rolling coefficient per specimen. Both have the same parameter count. No per-outcome adjustment or negative friction is used. The physical suggestion comes from viscoelastic theory, but these quasirolling tennis data alone cannot identify a true continuum relaxation time.

Specimen	Constant RMSE	Relaxation RMSE	Reduction
Moderately old tennis	0.002495	0.001209	51.5%
Old tennis	0.004042	0.001315	67.5%
New tennis	0.057620	0.038197	33.7%

RMSE is in the dimensionless effective rolling coefficient. Every one of the eight evaluation points has a smaller absolute residual; the maximum error decreases for each specimen. These points were not used to estimate the parameter, but the data and a first error preview were already inspected during model selection. This is **exploratory reuse of a split**, not new blind validation.



The horizontal black line is zero error. Crosses denote the constant comparator; dots denote the relaxation candidate. The left observable is rolling resistance, not a measured outgoing angular-velocity vector. The right contains all 25 release-height evaluation records for rocks; no unfavorable record is removed.

Contact-local mechanics with the user's impulse notation

Let τ_v denote relaxation time; the source denotes it by A . For a supported sphere of radius R , mass m , inertia $\mathbb{I} = \alpha m R^2$, normal load N , and pure rolling angular speed $\omega \geq 0$, the restricted viscoelastic law is

$$\mu_r = \tau_v \omega, \quad \delta L_s = -R \int \mu_r N dt, \quad \delta \mathbf{L} = \delta L_s \hat{\mathbf{s}}.$$

The independent angular impulse is essential. The body change includes the force lever arm as well as this independent impulse:

$$\Delta \mathbf{L} = \mathbf{r} \wedge \delta \mathbf{p} + \delta \mathbf{L}, \quad \mathbf{r} = -R \hat{\mathbf{n}}.$$

The supported no-slip condition is

$$\mathbf{v} + \mathbf{r} \omega = \mathbf{0}.$$

The right-hand wedge denotes the transpose; factor order is preserved. Resolving the static reaction together with the independent torque gives

$$\mathbb{I} \dot{\omega} + m R^2 \dot{\omega} = -\tau_v N R \omega, \quad \gamma = \frac{\tau_v N R}{\mathbb{I} + m R^2}.$$

For constant support load and geometry the exact finite update is

$$\omega^+ = \omega^- \exp(-\gamma h), \quad \delta \mathbf{L} = (\mathbb{I} + m R^2)(\omega^+ - \omega^-) \hat{\mathbf{s}}.$$

The parentheses here group a scalar sum. Matrix associativity does not allow removing grouping around sums. The linear impulse follows from $m \Delta \mathbf{v}$. It balances the force-induced angular change; it must not be omitted.

With $N = mg$, the instantaneous static demand satisfies

$$\frac{|F_t|}{N} = \frac{\tau_v \omega}{1 + \alpha} \leq \mu_s.$$

It is largest initially and decreases during this branch. The implementation rejects insufficient static capacity rather than asserting no slip. A sliding branch using μ_d is not implemented here. The state has no axial spin; it does not establish rolling/twisting closure for arbitrary 3D contacts.

The rigid kinetic energy is $E = \frac{1}{2}(\mathbb{I} + m R^2)\omega^2$, and the loss over a step is $E^- [1 - \exp(-2\gamma h)] \geq 0$. There is no resistive spin reversal. Small increments use `expm1` for numerical accuracy.

The scaled variables remain

$$\mathbf{V} = \begin{bmatrix} \mathbf{v} \\ \ell \omega \end{bmatrix}, \quad \mathbf{P} = \begin{bmatrix} \mathbf{p} \\ \mathbf{L}/\ell \end{bmatrix}, \quad \Delta \mathbf{P}_c = \begin{bmatrix} \delta \mathbf{p} \\ \delta \mathbf{L}/\ell \end{bmatrix}.$$

The reference length ℓ changes representation, not resistance or motion. At zero relative contact velocity the user's direction $\hat{\mathbf{t}}$ is undefined. This prototype obtains the static reaction from the rolling constraint; it does not silently redefine $\hat{\mathbf{t}}$ or claim to have completed that directional branch. This issue must be resolved when integrating contact history.

How indentation can determine normal restitution cheaply

Under a small-deformation spherical contact, a viscoelastic reduction is

$$F_n = \kappa x^{3/2} + \frac{3}{2}\tau_v \kappa \sqrt{x} \dot{x}, \quad m_{\text{eff}} \ddot{x} + F_n = 0.$$

Here x is compression, not an impulse symbol. The elastic energy and loss rate are

$$U = \frac{2}{5}\kappa x^{5/2}, \quad D_n = \frac{3}{2}\tau_v \kappa \sqrt{x} \dot{x}^2 \geq 0.$$

For incident normal speed u , the isolated collision reduces to one dimensionless parameter

$$\beta = \frac{3}{2}\tau_v \left(\frac{\kappa}{m_{\text{eff}}} \right)^{2/5} u^{1/5}.$$

The prototype integrates this reference problem once to build a monotone table. Runtime restitution can then be obtained by a cubic lookup. For profiles using a measured restitution at a reference condition, matching that characterization must remain an explicit calibration with known stiffness and mass. A single restitution value is not enough to identify all deformation properties.

Release occurs when the normal force first reaches zero during unloading. For $\tau_v > 0$ this happens before compression reaches zero. Extending the law until geometric recovery creates an attractive interval. The remaining spring energy is recorded separately as unresolved internal recovery. It is not erased or counted as proven heat. The isolated endpoint table does not track that internal state into a later impact. Endpoint restitution must not be imposed a second time on top of the resolved damping.

Efficient isolated pair maps are not simultaneous-contact solvers. Using one whole-impact map per pair in rapid groups can violate coupled momentum, work and contact history. A group implementation needs the coupled mobility solve already derived in the article, with local history and branch transitions.

The rock experiment rejects this simple viscoelastic extension

Both candidates fit one normal-response parameter using the same normalized normal-speed residual on 50 records. They evaluate on the existing 25 records released from 4.5 m. A further four-fold check leaves out complete slab-angle groups. The viscoelastic size law assumes homogeneous spherical mass and curvature: $\beta = b(0.05 \text{ m}/R)(u/(1 \text{ m/s}))^{1/5}$. The fitted b is an effective proxy, not independently identified viscosity.

Evaluation	Constant normal law	Viscoelastic sphere law
25 height records, normal RMSE (m/s)	1.00437	1.13969
75 pooled angle-fold records, normal RMSE (m/s)	1.14028	1.19421
Height-set maximum error (m/s)	2.29727	2.34718
Angle-fold maximum error (m/s)	2.33496	2.84656

The extension worsens both splits and their maximum errors. It is rejected for these rock records. Conditional tangential/spin calculations keep the historical fitted coefficients; these are not documented material values or actual-shape tests. Actual facet attitude and inertia remain unavailable. Failure does not show that all viscoelasticity is absent; it shows this particular reduction is insufficient.

A better cheap direction for rocks: local indentation geometry

The source experiment used faceted limestone samples and softer concrete slabs. It documents repeated surface damage, indentation edges and possible rapid second contacts that alter rebound direction. Its figure 12 labels these indentation dimensions:

Diameter (cm)	Depth (cm)	Simplified conical wall angle
3.5	1.5	40.60 degrees
1.7	1.6	62.02 degrees
2.3	0.6	27.55 degrees
2.1	0.5	25.46 degrees

These photographs are not matched to individual collision rows. Conical geometry is an illustrative reduction, not an inferred experimental contact frame. The depths are too large to assume the small Hertz half-space limit automatically.

A synthetic frictionless control makes the distinction clear. A rigid sphere arrives with $\mathbf{v}^- = [3, -1, 0]^T$ m/s, mass 1 kg and local $e_n = 0.5$. Against a horizontal normal, it leaves with $[3, 0.5, 0]^T$ m/s. Against a local normal $[-0.5, 0.86603, 0]^T$ tilted 30 degrees from the mean normal, it leaves with $[1.22548, 2.07356, 0]^T$ m/s. The ratio measured against the mean slab normal is 2.07356, while total kinetic energy decreases from 5 to 2.90072 J. The actual local restitution is still 0.5. Tangential incident energy is redirected upward; no negative friction or active elasticity is needed.

This control does not predict a real rock. A central sphere normal force has zero lever-arm torque. For facets, a pressure centroid displaced from the nominal contact point can also contribute an independent angular impulse. Distributed force and moment must be accumulated before checking the permitted directional components.

The next rock candidate should therefore retain a local indentation depth and rim/pressure geometry, use a documented or separately estimated yield/indentation law, and resolve a few contact sites. Local normals and pressure lever arms then follow geometry instead of a fitted function of the measured rebound. Bodies remain rigid. A few contact states add cost in proportion to contacts rather than body mesh nodes. Nominal C25 concrete strength is not an indentation-hardness measurement and cannot simply be substituted for one.

A better rubber bounce candidate: shear history plus pressure moment

Primary ball experiments show grip, tangential vibration and friction-force reversal. The rolling law above describes sustained support; it is not automatically a law for the short grip phase of a bounce. A small shear state and one or two independently characterized vibration modes can represent that phase. Static traction can return stored elastic energy while sliding friction opposes actual slip. Asymmetric normal pressure can supply the separate angular impulse.

For a practical benchmark, compare a force-history Mindlin reduction and a tiny contact-patch reference. Force history handles unloading more carefully than some displacement-rescaling variants, but energy accounting and frame transport must still be checked. Neither changing stiffness without accounting for stored energy nor clearing elastic history at separation is acceptable by itself. This continuation implements no new general shear/plastic/opening branch or empirical rubber-bounce fit.

Independent spin torque without a new fitted friction coefficient

An additional restricted patch calculation directly addresses the zero-center-slip spin case. For pure spin about the normal of a circular Hertz contact, the center has zero tangential velocity but the rest of the patch slides. With radial pressure $p(\rho) = p_0 \sqrt{1 - \rho^2/a^2}$, contact radius a , and the existing dynamic sliding coefficient μ_d , the integrated load and moment magnitude are

$$N = 2\pi \int_0^a p(\rho) \rho d\rho = \frac{2}{3}\pi p_0 a^2, \quad T = 2\pi \mu_d \int_0^a p(\rho) \rho^2 d\rho = \frac{\pi^2}{8} \mu_d p_0 a^3.$$

Consequently,

$$T = \frac{3\pi}{16} \mu_d N a, \quad \delta p_t = 0, \quad \delta \mathbf{L} = \delta L_n \hat{\mathbf{n}}.$$

The linear shear forces cancel by symmetry; their moment does not. The small impulse δL_n changes spin even though a center-point sliding rule would produce no tangential impulse. This is a standard traction integral, not a claim of new mechanics. For constant load and patch size with isotropic inertia, the ideal pure-sliding update followed by zero-external-torque arrest is

$$\delta L_n = -\text{sgn}(\omega_n) \min\left(\frac{3\pi}{16} \mu_d N a h, \mathbb{I}|\omega_n|\right), \quad \omega_n^+ = \omega_n^- + \frac{\delta L_n}{\mathbb{I}}.$$

The mechanical loss is $-\omega_n^- \delta L_n - \delta L_n^2/(2\mathbb{I}) \geq 0$. No extra spinning-friction coefficient is fitted. The physical length is contact radius a , not the coordinate reference length ℓ .

Twenty-four rotated spatial controls independently integrate distributed tractions and verify net force, torque, power and body kinetic energy. Zero friction and zero spin preserve the state; resistance arrests spin without reversal. For axial spin $\hat{\mathbf{s}}$ and $\hat{\mathbf{n}}$ are dependent; the prototype allocates this resultant to δL_n , not to two independently identifiable scalar couples. Mixed spin/translation, torsional elastic microslip and changing normal pressure are not resolved here. A transient's pressure distribution can differ from the Hertz profile, so this coefficient must not be transferred blindly to a rubber bounce. No measured spin improvement is claimed from these controls.

Literature comparators for the next coupled model

The pressure-field contact model of Elandt, Drumwright, Sherman and Ruina computes resultant force and moment from a contact patch while bodies stay nominally rigid. It is a useful reference for irregular geometry and independent moments. The pressure fields are prepared in advance; no body deformation field is solved at runtime. Its reported scope does not capture internal wave dynamics of shells, so a pressure field alone is insufficient for every rubber-ball vibration.

Drake provides an implementation reference. Its pressure modulus, resolution and damping need characterization; default values are not automatically documented material properties. This continuation did not run Drake or validate a pressure model against the collision data.

For local plastic indentation, Zunker and Kamrin's dimensionality-reduced elastic-perfectly-plastic contact models offer a mechanically derived comparison with contact history. They do not by themselves validate brittle rock/concrete crushing or recover missing facet orientation. Both references help constrain the next small patch/history model; neither is silently adopted as a material fit.

Physical verification and cost

The exact rolling update passes 100 planar and 100 rotated spatial controls at three representation lengths, including direct linear/angular momentum, kinetic energy, zero slip, time subdivision, zero resistance and static-capacity rejection. Twenty independent continuous-ODE comparisons pass. Maximum relative errors are about 2.04×10^{-15} for momentum and 1.15×10^{-15} for energy. This is a 3D sphere law restricted to planar motion when appropriate; it is not a derived plane-strain constitutive law for arbitrary 2D shapes.

Normal controls recover the analytic elastic collision duration and the known weak-damping coefficient. Twenty-five tighter reference controls keep energy balance error below 4.54×10^{-11} in dimensionless energy. The 513-node normal table covers $0 \leq \beta \leq 8$; 64 independent random tighter solves show maximum restitution error 1.35×10^{-7} . Out-of-range use is rejected. Grid arrays plus interpolation coefficients occupy 24,592 bytes, excluding language/object overhead. Local Python setup took about 7.75 seconds.

The C++ benchmark uses double precision, -O3 and no fast-math. Sixteen native/Python controls pass at rounding error; cached/uncached rolling checksums match. Median local kernel costs are:

Responses	Constant roll	Relaxation, exp	Cached factor	Normal map
100	4.18	12.89	2.81	31.25
10,000	4.06	12.87	2.71	31.29
1,000,000	8.56	13.56	7.78	31.42

All timings are nanoseconds per scalar response, including resultant impulse and rigid-energy changes. The normal kernel includes size/speed conversion and table lookup. The cached rolling factor requires unchanged support load, geometry and timestep; otherwise the exponential must be recomputed. One million response records occupy 88 MB of input/output arrays. These are synthetic array kernels, not one million interacting bodies or the native engine’s full scenes. Timings exclude detection, evolving frames, contact networks, branch transitions and simulation output. No all-case performance or 2x claim is made.

Parameters and evidence boundaries

Specimen	Estimated effective τ_v (ms)	Fit points	Evaluation points
Moderately old tennis	1.860	3	2
Old tennis	2.159	3	3
New tennis	25.039	3	3

Reported one-PDF-point sensitivities are about 1.826–1.895, 2.122–2.197 and 24.521–25.562 ms respectively. They are digitization sensitivity ranges, not confidence intervals. Leave-one-point-out diagnostics also favor the relaxation law for each specimen. The source uses belt-derived apparent speed and allows slight skid. It supplies no matched specimen mass, inertia, normal stiffness, impact restitution and static/dynamic friction characterization. Hence no claim that these fitted effective times can already predict the same balls’ bounces.

The production engine is unchanged. General full-wrench contact closure, native rolling/twisting, rapid irregular-group qualification and independently characterized full-state comparisons remain open. The numerical report’s prior failures remain available, and the user-removed 2x gate is not reinstated.

Sources and reproducibility

Brilliantov and Pöschel, *Rolling as a continuing collision*, EPJ B 12 (1999), 299–301; <https://arxiv.org/pdf/cond-mat/0203343>. Source rolling coefficient has dimensions of length: $M = \mu_{\text{roll}}N$ and $\mu_{\text{roll}} = \tau_v v$. Here $M = \mu_r RN$, so for pure rolling $\mu_r = \tau_v \omega$. Its scope requires small deformation and sufficiently slow quasi-static rolling relative to material relaxation; the tennis apparatus does not independently establish those assumptions.

Schwager and Pöschel, *Coefficient of restitution for viscoelastic spheres: the effect of delayed recovery*, Phys. Rev. E 78 (2008), 051304; <https://arxiv.org/pdf/0708.1434>. Repulsive force-zero release and delayed recovery motivate the normal reference stopping rule.

Cross, *Grip-slip behavior of a bouncing ball*, Am. J. Phys. 70 (2002), 1093–1102; <https://www.physics.usyd.edu.au/~cross/PUBLICATIONS/21.%20GripSlip.PDF>. Cross, *Impact of a ball on a surface with tangential compliance*, Am. J. Phys. 78 (2010), 716–720; <https://www.physics.usyd.edu.au/~cross/PUBLICATIONS/47.%20MoreSpin.pdf>. These sources support retaining contact shear/mode history rather than assuming all momentary zero slip is ordinary sustained rolling.

Wang et al., *Effects of the impact angle on the coefficient of restitution in rockfall analysis based on a medium-scale laboratory test*, NHESS 18 (2018), 3045–3061; <https://doi.org/10.5194/nhess-18-3045-2018>. Sections 2.2 and 5.1 and figure 12 document slab damage and proposed rebound constraints. Figure values are not per-impact calibration data.

Singh et al., 2008 tennis quasirolling source, <https://arxiv.org/pdf/0809.4823v2>. Experimental blue points from page 6 are reused from the separately archived full report; author regression curves are not the new validation targets.

LAMMPS granular model documentation, https://docs.lammps.org/pair_granular.html, accessed 7 October 2026. Its comparison of force/displacement unloading histories supplies an implementation reference, not independent proof of this project’s contact law.

Elandt et al., *A pressure field model for fast, robust approximation of net contact force and moment between nominally rigid objects*, IROS 2019; <https://arxiv.org/abs/1904.11433>. Drake Hydroelastic Contact User Guide, https://drake.mit.edu/doxygen_cxx/group__hydroelastic__user__guide.html. Zunker and Kamrin, *A mechanically-derived contact model for adhesive elastic-perfectly plastic particles*, parts I and II; <https://arxiv.org/abs/2309.07300> and <https://arxiv.org/abs/2309.07317>.

Reproduce from `research/viscoelastic-relaxation/` with NumPy, SciPy and Matplotlib installed. Run:

```
experiment.py --output NEW-DIRECTORY
audit.py --evidence NEW-DIRECTORY --output NEW-AUDIT
benchmark.py --evidence NEW-DIRECTORY --output NEW-BENCHMARK
```

Outputs refuse to overwrite existing evidence. `render.py` builds the authoritative scatter and report source; `build.sh` compiles this PDF. Machine-readable evidence contains all points, folds, retained failures and source fingerprints. Third-party PDFs are cached outside the repository.